

An association between dietary arachidonic acid, measured in adipose tissue, and ulcerative colitis.

BACKGROUND & AIMS: Dietary arachidonic acid, an n-6 polyunsaturated fatty acid (n-6 PUFA), might be involved in the etiology of ulcerative colitis (UC). We performed a prospective cohort study to determine whether high levels of arachidonic acid in adipose tissue samples (which reflects dietary intake) are associated with UC.

METHODS: We analyzed data collected from 57,053 men and women in the EPIC-Denmark Prospective Cohort Study from 1993 to 1997. Adipose tissue biopsy samples were collected from gluteal regions at the beginning of the study, the cohort was monitored over subsequent years, and participants who developed UC were identified. A subcohort of 2510 randomly selected participants were used as controls. Concentrations of arachidonic acid were measured in adipose tissue samples. In the analysis, arachidonic acid levels were divided into quartiles; relative risks (RR) were calculated and adjusted for smoking, use of aspirin and nonsteroidal anti-inflammatory drugs, and levels of n-3 PUFAs.

RESULTS: A total of 34 subjects (56% men) developed incident UC at a median age of 58.8 years (range, 50.0-69.0 years). Those in the highest quartile for arachidonic acid concentrations in adipose tissue had an RR for UC of 4.16 (95% confidence interval [CI]: 1.56-11.04); a trend per 0.1% increase in arachidonic acid of 1.77 in RR was observed (95% CI: 1.38-2.27). The fraction attributed the highest levels of arachidonic acid was 40.3%.

CONCLUSIONS: Individuals with the highest relative concentrations of arachidonic acid in adipose tissue have a significantly greater risk of developing UC. Dietary modifications might therefore prevent UC or reduce disease symptoms. Copyright © 2010 AGA Institute. Published by Elsevier Inc. All rights reserved.